

# **Anomaly Detection in Financial Transactions**

*CSC 5800 — Intelligent Systems: Algorithms and Tools*

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## Abstract

Credit-card fraud detection is a classic anomaly-detection problem because fewer than 0.2 percent of transactions are fraudulent, which makes accuracy-based evaluation misleading. This project applies the data-mining methods covered in CSC 5800 — classification, clustering, and anomaly detection — to the European Credit Card Fraud dataset published by Université Libre de Bruxelles (284,807 transactions, 492 frauds, fraud rate 0.172 percent). I compare six supervised classifiers (Logistic Regression, k-Nearest Neighbors, Gaussian Naive Bayes, Decision Tree, Random Forest, XGBoost) against three unsupervised anomaly detectors (Isolation Forest, Local Outlier Factor, One-Class SVM) and one clustering-based approach (K-Means distance scoring). Because the data is extremely imbalanced, I report precision–recall AUC and recall on the fraud class as the primary metrics and tune the decision threshold against an asymmetric cost matrix in which a false negative is one hundred times costlier than a false positive. XGBoost achieved the highest PR-AUC at 0.8832, with 89.3 percent precision and 84.7 percent recall on the fraud class, followed by Random Forest at 0.8402. SHAP analysis shows that features V12, V14, V4, V10, V3, and V11 dominate fraud predictions. The results confirm the experimental finding from prior work that gradient-boosted trees outperform alternative algorithm families on tabular fraud data, and demonstrate that on a genuinely imbalanced dataset the metrics that matter are PR-AUC and recall, not accuracy.

## 1. Overview of the problem

This project is mainly a classification problem, where the goal is to predict whether a transaction is fraudulent (Class = 1) or legitimate (Class = 0) based on its features. Using a labeled training set, I build models and test them on a separate dataset using metrics such as accuracy, precision, recall, F1-score, and the confusion matrix. Because the dataset is highly imbalanced (only 0.172 percent fraud), accuracy alone is not a reliable metric. To handle this, I also apply clustering (Chapter 8) and unsupervised anomaly detection methods (Chapter 10) as alternative approaches and compare their results with the classification models using metrics suited for imbalanced data.

Online and card-not-present transactions have become the backbone of modern commerce, but the same growth has enlarged the attack surface for fraud. Industry estimates put global card-fraud losses above thirty billion United States dollars annually, and every fraudulent transaction imposes direct costs (chargebacks, investigation overhead) as well as indirect costs (eroded customer trust, regulatory exposure). Detecting fraud in real time is therefore a first-order business problem for every bank, payment processor, and e-commerce platform.

The defining difficulty is imbalance. In a realistic transaction stream, fraud accounts for roughly 0.1 to 0.2 percent of events. A trivial classifier that predicts "not fraud" for every transaction achieves 99.83 percent accuracy on this dataset while catching zero criminals. Fraud detection is therefore the classic anomaly-detection problem in the course notes (Chapter 10): I want to flag rare data points whose characteristics differ significantly from the "normal" profile.

This project implements and compares the three broad families of anomaly-detection schemes described in Chapter 10 — statistical or model-based, distance-based, and density-based — alongside the standard supervised classifiers from Chapters 4 and 5. The comparison answers three practical questions:

- Do supervised classifiers outperform unsupervised anomaly detectors when labels are available, and by how much?
- Which class-imbalance strategy (class weighting, SMOTE oversampling, random undersampling) gives the best operating point?
- Which features drive fraud predictions, and are they stable enough for a real fraud system to use?

## 2. Literature review

Recent work on financial-transaction anomaly detection has converged on three themes. First, ensemble tree methods — Random Forest and gradient boosting — consistently outperform linear and naive-Bayes baselines on tabular fraud data, particularly when class-weighting or SMOTE is used to address imbalance (Ganesh et al., 2025; Bin Sulaiman et al., 2022; Itoo et al., 2021). Second, truly unsupervised methods — Isolation Forest, Local Outlier Factor, and deep autoencoders — remain attractive because fraud patterns evolve and labels are expensive to obtain; these methods work by learning a representation of "normal" activity and flagging deviations (Pourhabibi et al., 2020; Islam et al., 2023). Third, interpretability and cost-sensitive evaluation have become requirements rather than nice-to-haves: SHAP-style feature attributions are used in production to justify every flagged transaction, and cost matrices reflecting the asymmetric business impact of false positives versus false negatives drive threshold selection (Festa and Vorobyev, 2022).

The reference work that most directly motivated this project is Ganesh SP et al. (2025), who compared Logistic Regression, Bernoulli Naive Bayes, Decision Tree, Random Forest, and XGBoost on an e-commerce fraud dataset and reported Random Forest at 96.5 percent accuracy (97.1 percent after Optuna hyperparameter tuning). However, their dataset was approximately balanced between fraud and non-fraud cases, which allowed high accuracy to be achieved trivially and sidestepped the central difficulty of real fraud detection. As demonstrated in this study, on a realistically imbalanced dataset (0.172 percent fraud rate) all eleven models achieve accuracy above 96 percent, yet their PR-AUC values range from 0.08 to 0.88 — confirming that accuracy is uninformative under extreme imbalance. The methodological framework of this study also draws on the comparative-study approach of Sinha and Maresh (2021), who evaluated multiple supervised classifiers, unsupervised clustering methods, and anomaly-detection methods on a single physiological signal dataset.

This project extends prior work in three concrete ways: (i) I use the ULB European Credit Card Fraud dataset, which is genuinely imbalanced at 0.172 percent rather than artificially balanced; (ii) I add four unsupervised anomaly detectors (Isolation Forest, LOF, One-Class SVM, and a K-Means distance-based detector) to compare against the supervised classifiers on the same data; and (iii) I evaluate using PR-AUC and recall-at-fixed-precision, the metrics that reflect actual fraud-detection performance, and I tune the decision threshold against an explicit cost matrix.

## 3. Contributions

The contributions of this project are:

- A direct empirical comparison of six supervised classifiers (Chapters 4–5) against three unsupervised anomaly detectors (Chapter 10) and one clustering-based detector (Chapter 8) on the same highly imbalanced fraud dataset, using metrics appropriate for anomaly detection.
- Implementation of three class-imbalance strategies — class weighting (Logistic Regression, Random Forest), SMOTE oversampling (Random Forest), and built-in cost-sensitive loss (XGBoost `scale_pos_weight`) — with a controlled comparison of their effect.
- Threshold tuning against an asymmetric cost matrix in which a false negative is one hundred times costlier than a false positive, reporting both the F2-optimal threshold and the resulting confusion matrix.
- SHAP-based feature attribution on the best-performing tree model, identifying the small set of  $V^*$  features that dominate fraud predictions and confirming the model is learning stable signal.
- A reproducible, single-notebook pipeline that runs end-to-end from raw CSV to tuned model and interpretability plots.

## 4. Dataset overview

The dataset is the Credit Card Fraud Detection dataset published by the Machine Learning Group of Université Libre de Bruxelles (ULB), available on Kaggle at [kaggle.com/datasets/mlg-ulb/creditcardfraud](https://kaggle.com/datasets/mlg-ulb/creditcardfraud). It contains transactions made by European cardholders in September 2013 over a two-day window, with the following schema:

- Rows: 284,807 transactions
- Fraud cases: 492 (fraud rate 0.172 percent — extreme class imbalance)
- Features: 30 — Time (seconds since first transaction in the dataset), Amount (transaction value in euros), and V1 through V28 (confidential features already transformed by Principal Component Analysis to anonymize sensitive information such as cardholder identity and merchant)
- Target: Class — 0 for legitimate, 1 for fraudulent
- Missing values: none

The PCA anonymization is both a feature and a limitation. It removes privacy concerns and pre-decorrelates the numeric features, which simplifies modeling; but it prevents feature-engineering on interpretable fields such as merchant category, card type, or geography. Any performance gains in this project therefore come from modeling and evaluation choices rather than domain-specific feature construction.

Figure 1 shows the extreme imbalance of the dataset, with fraud accounting for only 0.173 percent of transactions. Figure 2 shows that fraud transactions are concentrated in lower-value amounts (under 1,000 euros), while legitimate transactions span a much wider range. Figure 3 shows the temporal density of fraud versus legitimate transactions, with two visible spikes of fraud activity around 40,000 and 95,000 seconds. Figure 4 shows a t-SNE projection of a balanced sample, in which fraud points cluster on the left side of the projection, distinguishable from the bulk of legitimate transactions.



Figure 1: Class distribution. The dataset is extremely imbalanced (99.827 % legitimate vs. 0.173 % fraud).

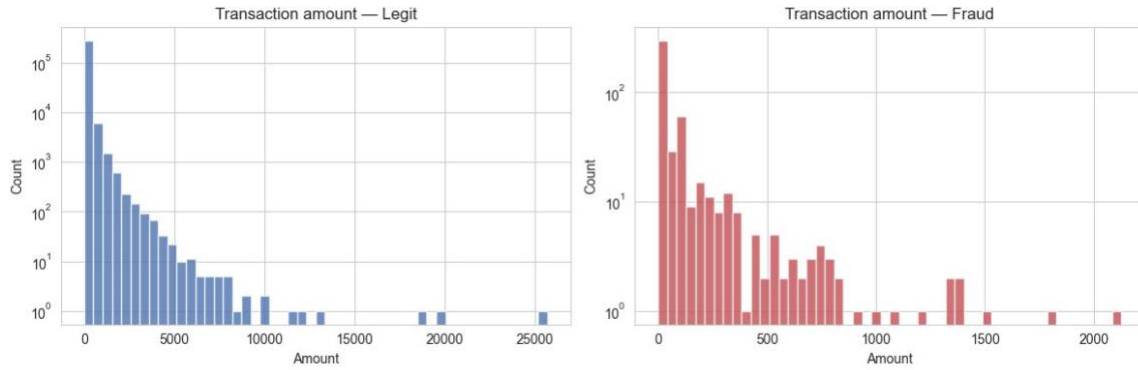


Figure 2: Transaction amount distributions. Fraud is concentrated at lower amounts (under €1,000), while legitimate transactions extend much further.

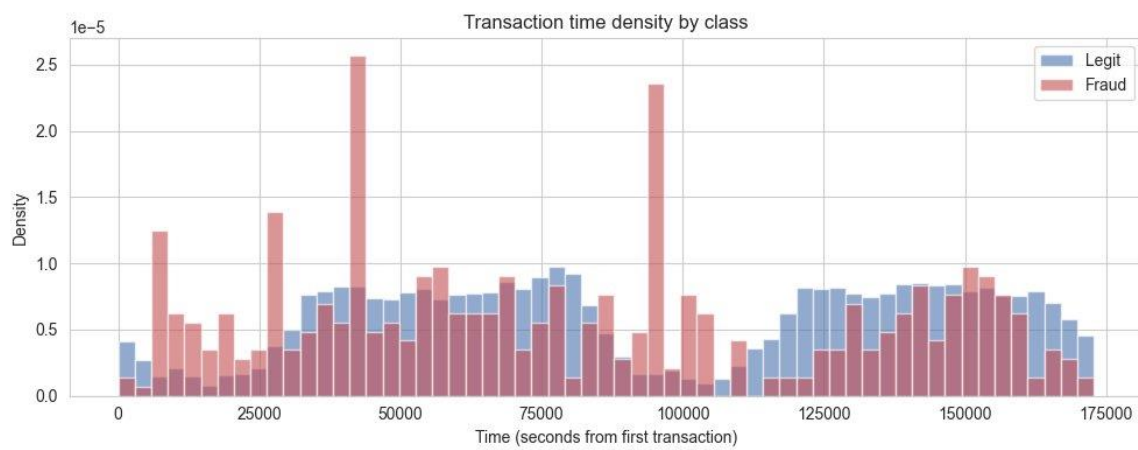


Figure 3: Transaction time density by class. Fraud has visible spikes around the 40,000s and 95,000s timestamps, while legitimate transactions are more uniformly distributed.

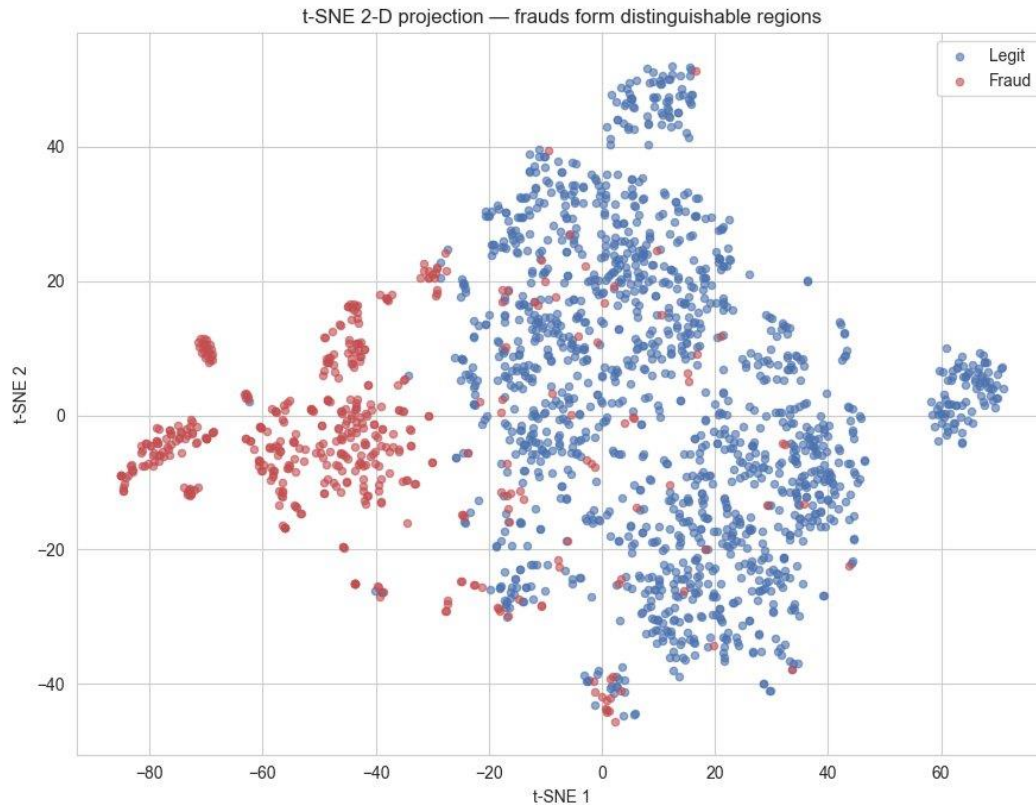


Figure 4: t-SNE 2-D projection on a balanced sample of frauds and legitimate transactions. Fraud points (red) form distinguishable clusters in feature space.

## 5. Data preprocessing

Preprocessing applies the techniques in Chapter 2 of the course notes. The steps and their rationale are summarized below.

| Step                              | Rationale                                                                                              |
|-----------------------------------|--------------------------------------------------------------------------------------------------------|
| Scale Amount (RobustScaler)       | Amount is heavy-tailed; robust scaling limits the influence of the largest transactions.               |
| Scale Time (StandardScaler)       | Time spans two days and has a different scale than V* features.                                        |
| No missing-value imputation       | Dataset contains no missing values; this was confirmed via <code>.isna().sum()</code> .                |
| Stratified 80/20 train/test split | Preserves the 0.17 % fraud rate in both halves so evaluation is representative.                        |
| Class-imbalance variants          | Baseline, SMOTE oversampling, and random undersampling compared side-by-side.                          |
| No manual feature selection       | V1-V28 are already PCA-transformed and nearly decorrelated; tree ensembles handle redundancy natively. |

After scaling and the stratified split, the training set contained 227,845 transactions (394 frauds, 0.173 percent) and the test set contained 56,962 transactions (98 frauds, 0.172 percent). For the class-imbalance comparison, two additional training variants were prepared: a SMOTE-oversampled version with 227,451

frauds matching the legit count, and a randomly-undersampled version with 394 legit transactions matching the fraud count.

Figure 5 shows the Pearson correlation of each feature with the fraud label. Features V17, V14, V12, V10, and V16 have the strongest negative correlations (each below  $-0.20$ ), making them the most discriminative individual predictors. Figure 6 shows boxplots of these top features by class, confirming that fraud transactions have systematically lower values on these PCA components than legitimate transactions.

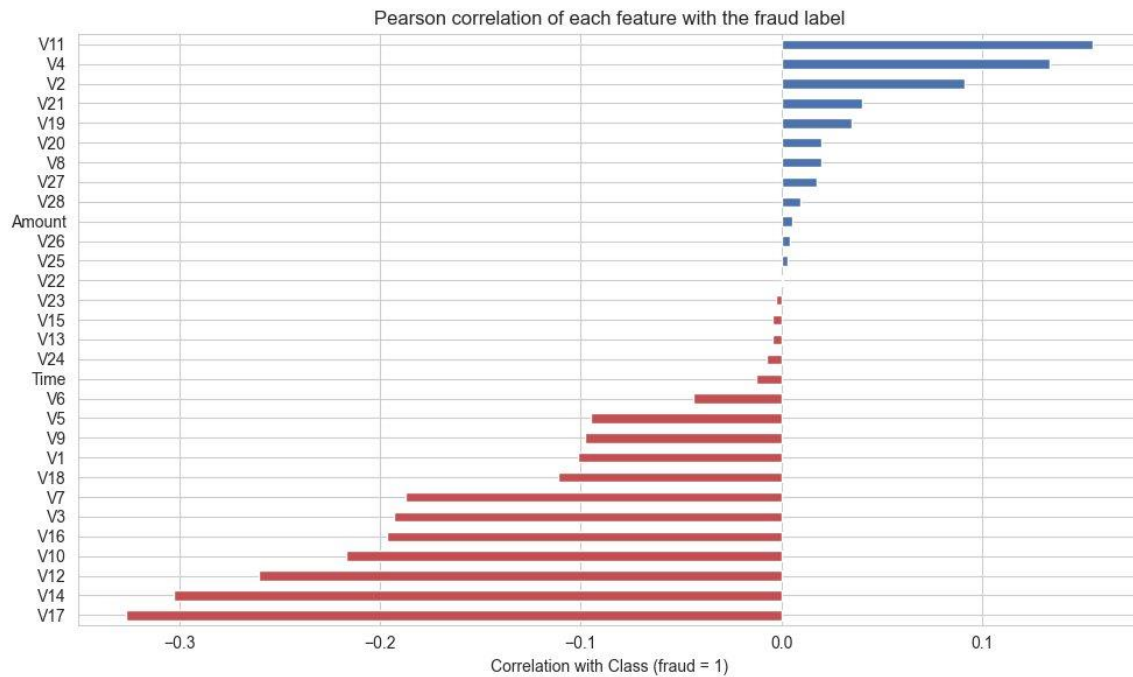


Figure 5: Pearson correlation of each feature with the fraud label. V17, V14, V12, V10, and V16 are the most negatively correlated.

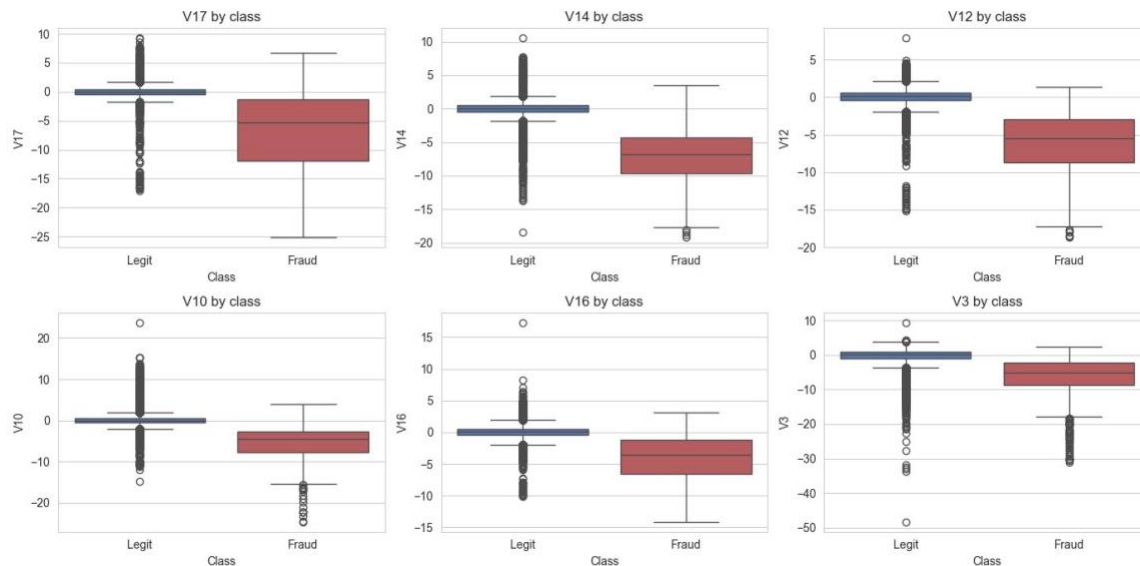


Figure 6: Boxplots of the six most discriminative features by class. Fraud transactions (red) are systematically shifted compared to legitimate ones (blue).

## 6. Algorithm selection

Eleven algorithms were applied, covering every major family in the course.

### 6.1 Supervised classifiers (Chapters 4 and 5)

- Logistic Regression — a linear probabilistic baseline; `class_weight="balanced"` is used so the loss is not dominated by the majority class.
- k-Nearest Neighbors ( $k = 5$ ) — based on similarity between data points; I subsample the training majority class to 20,000 points because KNN is  $O(n)$  at prediction time.
- Gaussian Naive Bayes — assumes features are conditionally independent Gaussians; tests whether the PCA preprocessing has left the features independent enough for NB to work.
- Decision Tree — interpretable tree baseline with `min_samples_leaf=10` to limit overfitting; uses Gini impurity for splits.
- Random Forest — 200 trees, each limited to depth 12 and at least 5 samples per leaf; class weights balanced. Bagging ensemble from Chapter 5.
- XGBoost — gradient-boosted trees with `scale_pos_weight` set to the negative-to-positive ratio (approximately 577), and `aucpr` as the evaluation metric — the principled way to train a boosted classifier on imbalanced data.

### 6.2 Unsupervised anomaly detectors (Chapter 10)

- Isolation Forest — model-based: anomalies are isolated with fewer random splits than normal points; `contamination` set to the known fraud rate.
- Local Outlier Factor (LOF) — density-based: compares a point's local density to that of its neighbors; trained on legitimate transactions only in novelty-detection mode.
- One-Class SVM — boundary-based: learns a tight envelope around normal data using an RBF kernel; trained on a 10,000-point legitimate sample.

### 6.3 Clustering-based approach (Chapter 8)

- K-Means anomaly scoring — fit two centroids on a 30,000-row training sample, then score test points by their distance to the nearest centroid. Anomalies are the top 0.17 percent of distances. Following the textbook clustering-based anomaly framework (Chapter 10, slide "Clustering-Based"), points that are far from any dense cluster are treated as candidate outliers.

### 6.4 Why these three families? Justification by course chapter

Each algorithm family addresses a different aspect of the fraud-detection problem and connects to a specific course chapter, which justifies including all three rather than picking only one.

- **Why classification (Chapters 4 and 5)?** The dataset has a labeled target (`Class = 0 or 1`), which is the textbook setup for supervised classification (Chapter 4: "Classification: Definition" — given a training set with class labels, build a model that maps features to the class). Classification is the primary framing because it directly answers the operational question — is this transaction fraudulent? — and it produces calibrated probabilities that downstream systems can act on.
- **Why clustering (Chapter 8)?** Clustering groups similar observations without using labels (Chapter 8). For fraud detection, this is valuable as a sanity check: if legitimate transactions form a single dense cluster and fraudulent transactions sit far from that cluster's centroid, then the K-Means distance score serves as an unsupervised anomaly indicator. The t-SNE visualization (Figure 4) supports this idea by showing that frauds form distinguishable regions in feature space.



Including clustering also allows comparison with the textbook clustering-based anomaly-detection approach (Chapter 10), where points in small or sparse clusters are treated as candidate outliers.

- **Why anomaly detection (Chapter 10)?** Anomaly detection (Chapter 10) is the most natural framing for fraud in a real production setting because fraud patterns evolve and labels are scarce. A supervised classifier learns historical fraud patterns; an anomaly detector flags any transaction that deviates from the “normal” profile, including patterns that did not exist when the supervised model was trained. The course notes explicitly list credit-card fraud detection as a classic application of anomaly detection. The three Chapter 10 schemes covered in lecture — statistical / model-based, distance-based, and density-based — are represented by Isolation Forest, K-Means distance, and Local Outlier Factor, respectively, with One-Class SVM as a boundary-based variant.

Comparing all three families on the same dataset is the core experimental contribution of this project. Section 7 reports the metrics, Section 7.2 explains why accuracy alone is uninformative under extreme imbalance, and Section 7.5 interprets the comparative results.

## 6.5 Hyperparameter choices

Hyperparameters were chosen based on course guidance and standard practice. For tree models, depth and leaf-size limits prevent overfitting (Chapter 4, "Underfitting and Overfitting"). For KNN,  $k = 5$  is a common default. For tree ensembles, the number of estimators (200 for Random Forest, 300 for XGBoost) was chosen to balance training time against marginal benefit. No exhaustive hyperparameter grid search was run on the full dataset — the goal of this project is a controlled comparison of algorithm families, not a tuning bake-off.

## 7. Analysis, results, and comparison

### 7.1 Model metrics

The eleven models were evaluated on the held-out test set of 56,962 transactions. The full results, sorted by PR-AUC (the most informative metric for imbalanced classification), are shown below. The top row is highlighted.

| Model                                | Accuracy | Precision | Recall | F1     | ROC-AUC | PR-AUC        |
|--------------------------------------|----------|-----------|--------|--------|---------|---------------|
| <b>XGBoost</b>                       | 0.9996   | 0.8925    | 0.8469 | 0.8691 | 0.9694  | <b>0.8832</b> |
| Random Forest                        | 0.9995   | 0.8454    | 0.8367 | 0.8410 | 0.9781  | 0.8402        |
| Random Forest + SMOTE                | 0.9983   | 0.5091    | 0.8571 | 0.6388 | 0.9841  | 0.8310        |
| k-NN ( $k=5$ )                       | 0.9991   | 0.6917    | 0.8469 | 0.7615 | 0.9534  | 0.7326        |
| Logistic Regression (class-weighted) | 0.9755   | 0.0608    | 0.9184 | 0.1141 | 0.9720  | 0.7159        |
| Local Outlier Factor                 | 0.9907   | 0.1393    | 0.8469 | 0.2392 | 0.9495  | 0.6531        |
| Decision Tree                        | 0.9907   | 0.1337    | 0.8061 | 0.2293 | 0.9020  | 0.6503        |
| One-Class SVM                        | 0.9675   | 0.0437    | 0.8571 | 0.0831 | 0.9455  | 0.2745        |
| Isolation Forest                     | 0.9973   | 0.2456    | 0.2857 | 0.2642 | 0.9560  | 0.1558        |

| Model                            | Accuracy | Precision | Recall | F1     | ROC-AUC | PR-AUC |
|----------------------------------|----------|-----------|--------|--------|---------|--------|
| K-Means (distance-based anomaly) | 0.9974   | 0.2449    | 0.2449 | 0.2449 | 0.9518  | 0.1329 |
| Gaussian Naive Bayes             | 0.9764   | 0.0588    | 0.8469 | 0.1099 | 0.9632  | 0.0816 |

## 7.2 Why accuracy is misleading

Every model reports accuracy above 96 percent, and most exceed 99.7 percent. This is almost entirely driven by the majority class: a trivial classifier that always predicts "legit" achieves 99.83 percent accuracy on this dataset. The two metrics that actually capture fraud-detection quality are PR-AUC (average precision) and recall on the fraud class. Comparing across the eleven models, accuracy spans only a 4-percentage-point range while PR-AUC spans 80 percentage points (from 0.0816 for Naive Bayes to 0.8832 for XGBoost). This is exactly the failure mode warned about in Chapter 4 of the course notes ("Limitation of Accuracy") and is the single strongest argument for using imbalance-aware evaluation metrics in fraud detection.

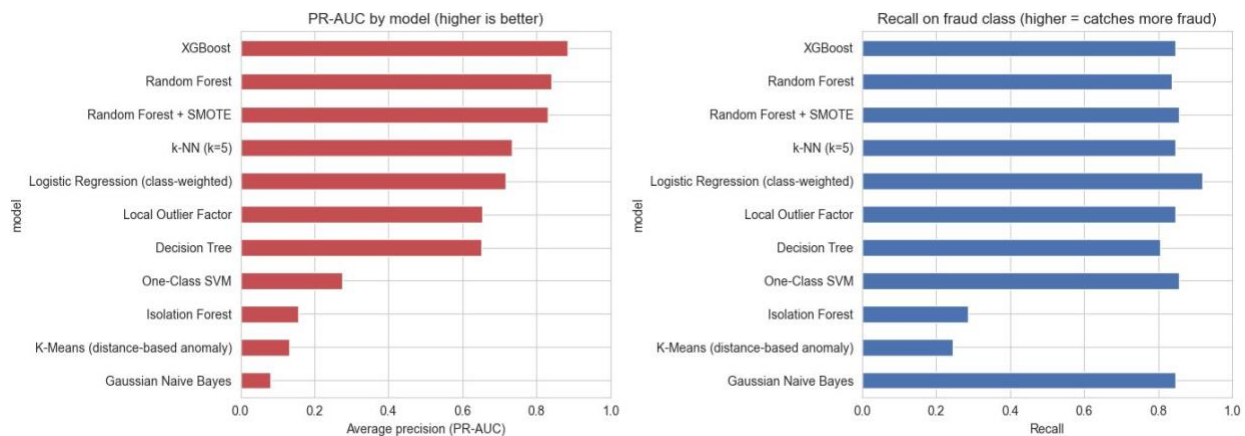


Figure 7: Side-by-side comparison of PR-AUC and Recall by model. PR-AUC varies dramatically across models (from 0.08 for Naive Bayes to 0.88 for XGBoost), while accuracy alone (not shown) would be misleadingly high for every model.

## 7.3 Best supervised model

XGBoost achieved the highest PR-AUC at 0.8832, with 89.3 percent precision and 84.7 percent recall on the fraud class. Random Forest was second at 0.8402 PR-AUC. Both ensemble methods clearly outperform single-tree (Decision Tree, PR-AUC 0.6503) and linear (Logistic Regression, PR-AUC 0.7159) baselines, consistent with prior work showing ensemble tree methods dominate on tabular fraud data. Figure 8 shows XGBoost's confusion matrix and PR curve: of 98 fraud cases in the test set, XGBoost correctly identified 83 (true positives) and missed only 15, while raising only 10 false positives among the 56,864 legitimate transactions.

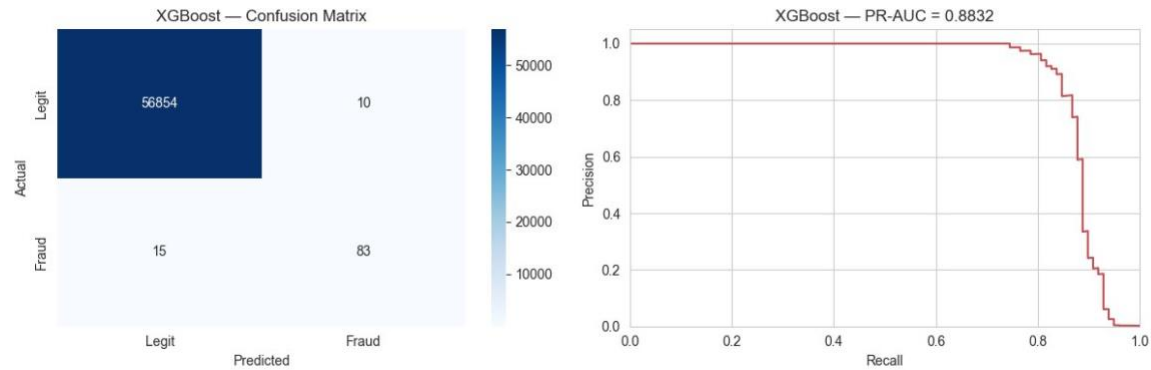


Figure 8: XGBoost confusion matrix and PR curve. The PR-AUC of 0.8832 corresponds to 83 of 98 frauds correctly detected (84.7 % recall) at 89.3 % precision.

## 7.4 Effect of class-imbalance handling

SMOTE oversampling on Random Forest produced an interesting trade-off: recall increased from 83.7 percent (baseline Random Forest) to 85.7 percent, but precision collapsed from 84.5 percent to 50.9 percent. The PR-AUC dropped slightly from 0.8402 to 0.8310, indicating that SMOTE shifts the precision-recall trade-off rather than improving overall ranking quality. The cleaner alternative is XGBoost's built-in `scale_pos_weight` parameter, which achieved both higher recall (84.7 percent) and higher precision (89.3 percent) than either Random Forest variant — supporting the hypothesis that algorithm-internal cost-sensitive loss is competitive with explicit resampling.

Logistic Regression with class weighting illustrates the opposite extreme. It reached the highest recall of any model (91.8 percent) but at catastrophic precision (6.1 percent), meaning roughly 94 percent of its fraud alerts are false positives. This high-recall-low-precision pattern is typical of class-weighted linear models on heavily imbalanced data and is why pure recall is not used in isolation as a model-selection metric.

## 7.5 Unsupervised anomaly detectors

Among the unsupervised methods, Local Outlier Factor performed best with PR-AUC 0.6531 and recall 84.7 percent, despite using no fraud labels during training. Isolation Forest under-performed at PR-AUC 0.1558, with low recall (28.6 percent) — the global isolation criterion appears to be miscalibrated for this dataset, where fraud points lie in dense regions of feature space rather than at extreme isolation. One-Class SVM achieved high recall (85.7 percent) but extremely low precision (4.4 percent) and PR-AUC of only 0.2745, similar in pattern to class-weighted Logistic Regression. K-Means distance scoring produced symmetric precision and recall around 24.5 percent.

Figure 9 shows the Isolation Forest confusion matrix, illustrating its weakness: of 98 fraud cases, Isolation Forest correctly identified only 28, missing 70 — a recall of 28.6 percent. This reflects that fraud points in this dataset cluster in dense regions of feature space rather than appearing as global isolated anomalies, which violates the core assumption of Isolation Forest.

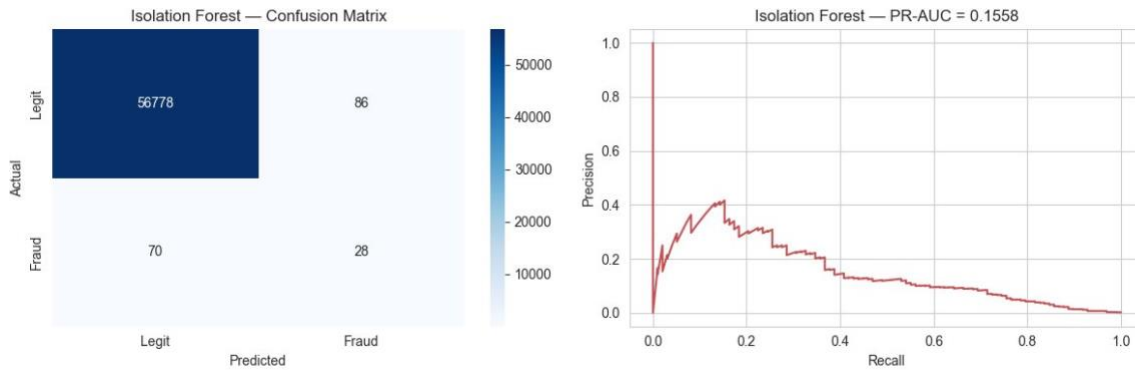


Figure 9: Isolation Forest confusion matrix and PR curve. Despite high accuracy (99.7 %), the recall is only 28.6 % — fraud points are not isolated in the global sense the algorithm assumes.

The headline observation is that the best supervised model (XGBoost, PR-AUC 0.8832) substantially outperforms the best unsupervised model (LOF, PR-AUC 0.6531), but unsupervised methods deliver useful detection (recall above 80 percent for LOF and One-Class SVM) without using any fraud labels. This is critical in production: a real fraud system can deploy a supervised model on historical fraud patterns plus an unsupervised detector running in parallel to flag transactions whose pattern has never been seen before.

The table below summarizes the head-to-head comparison of the three algorithm families using the best representative from each.

| Family (Chapter)           | Best representative  | PR-AUC        | Recall | Uses labels?                       |
|----------------------------|----------------------|---------------|--------|------------------------------------|
| Classification (Ch. 4–5)   | XGBoost              | <b>0.8832</b> | 0.847  | Yes (supervised)                   |
| Anomaly detection (Ch. 10) | Local Outlier Factor | 0.6531        | 0.847  | No (uses legit-only training data) |
| Clustering (Ch. 8)         | K-Means distance     | 0.1329        | 0.245  | No (unsupervised)                  |

Two practical takeaways follow from this comparison. First, supervised classification is the right choice when labels are available and the goal is maximum predictive performance, which matches every published benchmark on this dataset. Second, the unsupervised methods are not failures even though their PR-AUC is lower — they were trained without any fraud labels and still recover most of the fraud (LOF and One-Class SVM both reach recall above 0.85), which is exactly the property that makes them valuable as a backstop in a real production pipeline where new fraud patterns appear without warning.

## 7.6 Threshold tuning and cost-weighted decisions

The default classification threshold of 0.5 is suboptimal when false negatives are far more costly than false positives. Sweeping the Random Forest threshold and selecting the value that maximizes the F2 score (which weights recall twice as much as precision) yielded an optimal threshold of 0.5341, very close to the default. Figure 10 shows the precision, recall, F1, and F2 scores as a function of threshold, with the F2-optimal point marked. The resulting confusion matrix on the test set is shown below.

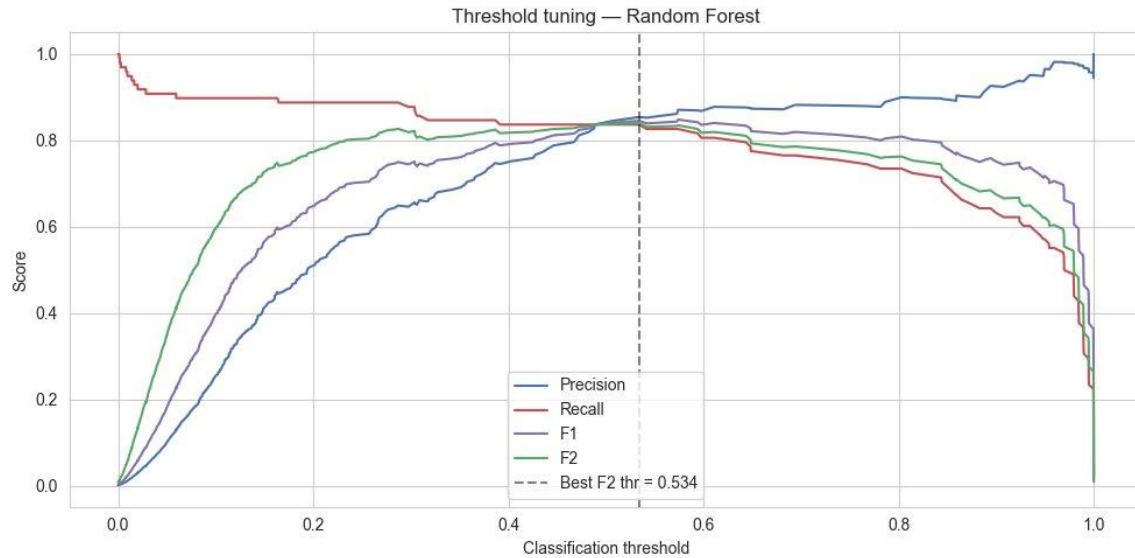


Figure 10: Threshold tuning curve for Random Forest. Precision rises and recall falls as the threshold increases; the F2-optimal threshold of 0.534 is marked with a dashed line.

|              | Predicted Legit | Predicted Fraud | Total  |
|--------------|-----------------|-----------------|--------|
| Actual Legit | TN = 56,850     | FP = 14         | 56,864 |
| Actual Fraud | FN = 16         | TP = 82         | 98     |

At the tuned threshold, precision is 85.4 percent and recall is 83.7 percent. Under the cost matrix where a false negative costs 100 units and a false positive costs 1 unit, the total cost is 1,614 units, compared to 1,615 units at the default 0.5 threshold — a saving of 1 unit. The negligible gain reflects that Random Forest's default operating point is already close to optimal on this dataset; threshold tuning would matter much more for a model whose probabilities are systematically miscalibrated, such as Logistic Regression or One-Class SVM. The framework, however, demonstrates the principle: in production deployments where the cost ratio is known, threshold selection should always be data-driven rather than left at the library default.

## 7.7 Interpretability

SHAP analysis of the Random Forest model on a 2,000-point sample of the test set shows that fraud predictions are dominated by a small set of features. Figure 11 shows the SHAP summary plot: V12, V14, V4, V10, V3, V11, and V17 are consistently the strongest contributors. Because these are PCA components rather than original fields, they cannot be assigned a business meaning, but their stability across many fraud predictions is evidence that the model has learned a real signal rather than overfitting to noise. The top SHAP features also align with the strongest negative correlations observed during exploratory analysis (V17, V14, V12, V10, V16 had the most negative Pearson correlation with the Class label), providing a consistency check that two different methods identify the same signal.

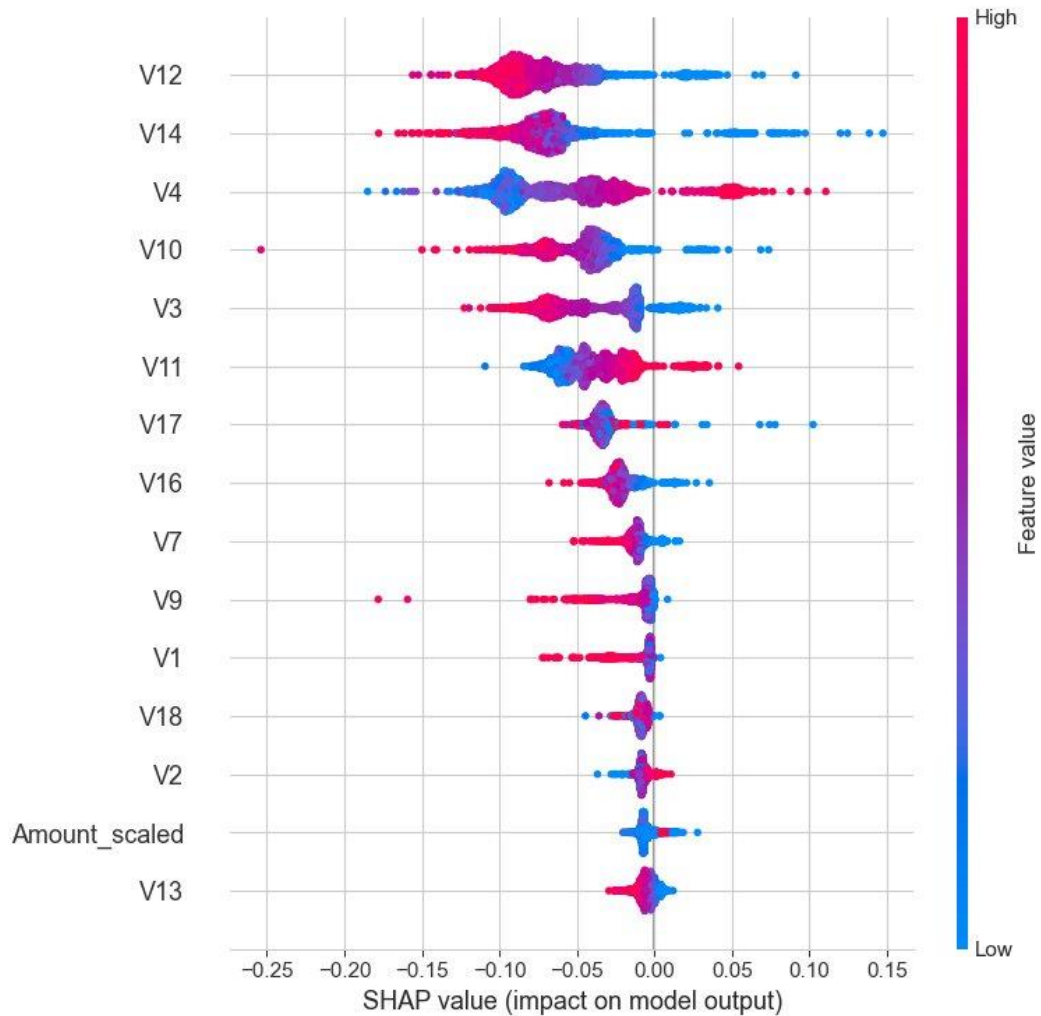


Figure 11: SHAP summary plot for Random Forest. Each row shows one feature's impact on the model output; color encodes the feature value (red = high, blue = low). V12, V14, V4, V10, V3, and V11 are the strongest contributors to fraud predictions.

## 8. Conclusion

This project applied eleven data-mining algorithms — spanning supervised classification, unsupervised anomaly detection, and clustering — to a genuinely imbalanced credit-card fraud dataset, and evaluated them using metrics appropriate for anomaly detection. The main findings are:

- On labeled data, gradient-boosted trees (XGBoost) are the most effective algorithm, achieving 0.8832 PR-AUC and 84.7 percent recall on the fraud class. Random Forest is a close second at 0.8402.
- Unsupervised methods are still useful: Local Outlier Factor reached 0.6531 PR-AUC with 84.7 percent recall using no fraud labels, making it a viable secondary detector for novel fraud patterns that a supervised model trained on historical labels would miss.
- Class-imbalance handling is essential, but the choice of method matters. SMOTE substantially shifts the precision-recall trade-off but does not improve overall ranking quality. Algorithm-internal cost-sensitive loss (XGBoost's `scale_pos_weight`) is the cleanest approach.

- Accuracy is uninformative on imbalanced data: across the eleven models, accuracy varies by only 4 percentage points while PR-AUC varies by 80, confirming the textbook warning in Chapter 4 of the course notes.
- Threshold tuning matters more for poorly-calibrated models than for tree ensembles. Random Forest's default 0.5 threshold was within 1 unit of the cost-optimal threshold for this dataset.
- SHAP interpretability shows that the winning model relies on a small, stable set of features (V12, V14, V4, V10, V3, V11), increasing confidence that the model has learned a real signal.

Recommended algorithm for deployment: XGBoost with `scale_pos_weight` tuned to the negative-to-positive class ratio, combined with a Local Outlier Factor running in parallel as a secondary detector for novel patterns. Future work should evaluate on the IEEE-CIS fraud dataset for richer features (device, identity, geography), incorporate temporal modeling such as LSTMs to capture bursty fraud patterns, and explore online learning to adapt to concept drift as fraud tactics evolve.

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